# VENUGOPAL K

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## SUMMARY

Final-year Computer Science student with a strong interest in Artificial Intelligence, Machine Learning, and Edge AI. Hands-on experience in Python, machine learning, computer vision, data preprocessing, and SQL. Experienced in developing predictive models and real-time object detection applications, with strong analytical and problem-solving skills.

## EDUCATION

**SRM University, Chennai, Tamil Nadu** Aug 2023 – Aug 2027

Bachelor of Technology in Computer Science **CGPA: 8.45/10**

## EXPERIENCE

**MSPL Limited, Hospet – Computer Vision Intern** Dec 2025 – Jan 2026

GitHub: Smart Employee Safety Management

- Developed an AI-powered Smart Employee Vehicle Safety & Access Control System using YOLO and OCR.
- Implemented real-time helmet detection and number plate recognition for industrial safety compliance.
- Contributed to computer vision and real-time object detection solutions for enterprise applications.

## PROJECTS

**AgriXAI – Empirical Study of Crop Yield Prediction** Feb 2026 – Apr 2026

*Python, Scikit-learn, Pandas, SHAP* Live Project

- Built a crop yield prediction model using 19,689 agricultural records.
- Compared 10 machine learning models using $R^2$, MAE, and RMSE for performance evaluation.
- Achieved 98.75% $R^2$ with Random Forest and applied SHAP for model explainability.

**Kidney Stone Prediction Using Machine Learning** Oct 2025 – Dec 2025

*Python, Scikit-learn, Pandas, Matplotlib, Seaborn* GitHub

- Built a risk prediction model using 4,000 patient records with 23 features.
- Performed EDA, data preprocessing, feature selection, and comparative model evaluation.
- Achieved 94.46% accuracy using Extra Trees Classifier.

## TECHNICAL SKILLS

**Programming:** Python, C++, SQL

**AI/ML:** Machine Learning, Deep Learning Fundamentals, Scikit-learn, Pandas, NumPy

**Computer Vision:** OpenCV, YOLO, OCR, Object Detection

**Data & Analytics:** Exploratory Data Analysis, Data Preprocessing, Feature Engineering, Model Evaluation

**Databases & Tools:** MySQL, PostgreSQL, Git, GitHub, Docker, Postman

**CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## CERTIFICATIONS

**SAP:** Certified Data Analyst – SAP Analytics Cloud

**Tata iQ:** GenAI Powered Data Analytics | Forage

**NPTEL:** Machine Learning | **Scaler:** Database Management System (DBMS)

## ACHIEVEMENTS

**Research:** AgriXAI paper shortlisted for Elsevier SSRN Reputed Series (Top 15%), ICICC 2027.

**Hackathons:** Hacktrax Hackathon 2025; Adobe India Hackathon 2024.